# Water Quality Across the United States

## Topic Overview

Water quality influences everyday life in ways that we rarely think twice about. Every community relies on a local water system, and each system operates under different conditions depending on its location, population, and water source. When incidents of water contamination occur from bacteria, disinfection byproducts, or aging infrastructure, they can provide important insight into environmental and community conditions[1]. As a result, these contamination events should be taken seriously, helping us understand how to prevent future cases.

The water systems that ensure water quality vary widely in size, funding, age, and environmental conditions. Large cities often rely on surface reservoirs and complex treatment plants, while small towns often depend on groundwater sources, which may face different contamination risks[2]. Some systems test and upgrade frequently, while others face aging infrastructure. Meanwhile, extreme weather continues to disrupt this infrastructure, making water quality an increasingly important environmental consideration.

This project gives you the chance to explore Environmental Protection Agency (EPA) data from thousands of water systems. You will analyze how violations, contaminants, water sources, and population size vary across ZIP codes. By studying these patterns, you will learn how environmental conditions shape community health and how data can help identify risks, inequities, and opportunities for improvement.

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1 https://www.epa.gov/dwreginfo/drinking-water-regulations
2 https://www.epa.gov/ground-water-and-drinking-water

## Dataset Description

This dataset is sourced from the U.S. EPA, which oversees national drinking water standards and monitors compliance across public water systems. It contains records for thousands of water systems across the United States, each mapped to its ZIP code. There are 41,345 rows, each representing one water system in one ZIP code.

The dataset includes 15 variables capturing key aspects of each water system.

All values come from EPA compliance records, which are based on laboratory testing, routine monitoring, and regulatory reporting. The dataset has been cleaned to remove incomplete or irrelevant fields from the original EPA data.

**Access the dataset here.** Review the dataset documentation here.

## Potential Questions

**How does water quality vary across geographic regions, and what environmental or infrastructural factors might explain the spatial patterns we see?**

*Plot the water systems on a map using their latitude and longitude. Customize the points so that color represents water source (i.e., groundwater vs. surface water), size represents total violations, and opacity or shape indicates whether the system has active issues. After creating the visualization, describe any geographic patterns you notice (e.g., clusters, regional trends, etc.). Then test a hypothesis about these patterns. For example, compare the average number of violations between groundwater and surface water systems using a two-sample t-test. Interpret the test statistic and p-value in the context of the map.*

**Do certain regions of the United States experience systematically different water quality outcomes, and what variables best explain those differences?**

*Divide the dataset into regions using ZIP-code prefixes or a region lookup. Find these statistics for each region: average total violations, contaminant counts, proportion of systems with active issues, and distribution of water sources. Which regions have the highest violation rates or most groundwater systems? Build a linear regression model to explore whether population served, water source, or region predicts total violations. Examine the coefficients and significance levels. What do the results suggest about how infrastructure or environmental conditions vary across regions?*

## Topics to Learn More About

- **Global Water Quality Rankings -** *It is important to get a global perspective on how countries differ in water quality, infrastructure, and environmental pressures. Understanding these patterns help you contextualize the ZIP code level water data you analyze in the project.*
  - **Water Quality by Country 2026** - Pay attention to which countries struggle with contamination, which excel and how geography affects water safety. These themes mirror similar factors in U.S. water systems.
- **Importance of drinking water -** *Safe drinking water is essential for human health, and even small changes in water quality can lead to serious illness or community impacts, which why the EPA does such rigorous water testing.*
  - **Drinking-water** - This WHO page explains the global health risks associated with unsafe drinking water and explains major causes of contamination. Read this to understand why water quality is so important.

## Additional Notes

- Some water systems have zero violations or zero contaminants simply because they have not been tested recently. Encourage students to think about what the zero means in the dataset.
- ZIP code coverage is not uniformly distributed across states; remind students of this when comparing regions and looking at national patterns.
- Turn categorical variables into numeric values. Water_source and has_active_issues will need to be encoded for regression or KNN (e.g. True = 1, False = 0, groundwater = 1, surface water = 0).

## Advanced Research

**Can we predict which water systems are at risk for active issues using violation history, contaminants, population, and water source characteristics?**
*Use the dataset to build a predictive model for whether a water system has active issues. Select explanatory variables such as total violations, health violations, unresolved violations, contaminant count, population, and water source. Train a K-Nearest Neighbors classifier and experiment with different values of k. Finally, evaluate the model. Then compare KNN to a logistic regression model by predicting the same outcome.*